

IRBID, Jordan

WMO: 402550

Lat: 32.550N

Lon: 35.850E

Elev: 619

StdP: 94.11

Time Zone: 2.00 (JOR)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.8	3.4	-7.6	2.1	11.4	-5.2	2.6	12.4	9.4	8.5	8.4	9.1	1.7	130	0.371

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	10.3	34.8	19.4	33.1	19.4	31.9	19.3	23.3	28.9	22.5	27.7	21.8	26.8	3.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.9	17.9	25.4	21.1	17.0	24.5	20.4	16.3	23.9	73.2	29.0	69.7	27.5	67.0	27.2	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.3	6.4	-0.6	38.7	0.9	1.8	-1.3	40.0	-1.8	41.1	-2.3	42.1	-3.0	43.4		
			-1.8	24.2	1.4	1.3	-2.8	25.2	-3.6	26.0	-4.4	26.7	-5.4	27.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	9.1	10.5	13.4	17.0	21.8	24.1	25.8	26.1	24.6	21.2	15.5	11.2
	DBStd	6.74	2.74	2.91	3.64	4.39	3.59	2.63	1.81	1.85	2.12	3.02	3.01	2.96
	HDD10.0	104	49	24	8	2	0	0	0	0	0	0	2	19
	HDD18.3	1073	287	219	161	76	8	0	0	0	0	9	92	221
	CDD10.0	3172	21	39	112	213	365	424	489	500	437	346	168	57
	CDD18.3	1098	0	1	7	37	114	174	231	242	187	97	8	1
	CDH23.3	9001	0	2	49	352	1063	1525	1952	2036	1379	598	41	2
	CDH26.7	3389	0	0	9	111	413	596	767	830	487	174	3	0
Wind	WSAvg	2.4	2.0	2.1	2.5	2.4	2.4	3.1	3.5	3.2	2.5	1.6	1.5	1.7
Precipitation	PrecAvg	483	99	92	90	26	6	1	0	0	1	15	51	94
	PrecMax	676	270	246	215	193	38	18	0	0	8	52	214	251
	PrecMin	267	18	22	4	0	0	0	0	0	0	0	0	16
	PrecStd	102	57	56	46	34	10	3	0	0	2	14	45	63
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.4	22.9	27.4	32.6	36.5	35.8	35.7	36.7	36.1	33.2	27.1	22.6
		MCWB	11.0	13.2	17.0	17.6	18.4	18.6	21.1	20.3	18.9	18.4	16.1	13.7
	2%	DB	16.9	19.8	24.3	29.6	33.1	33.9	33.2	33.5	33.0	30.6	24.8	19.5
		MCWB	10.0	12.1	13.7	16.1	17.4	19.3	20.4	20.6	18.8	18.0	15.0	12.0
	5%	DB	15.1	17.4	22.1	27.2	30.9	31.8	31.8	32.0	30.9	28.5	22.8	17.3
		MCWB	9.4	10.5	12.8	15.2	16.9	18.7	19.8	20.7	18.8	17.5	14.2	10.9
	10%	DB	13.6	15.5	19.9	24.9	28.7	30.1	30.7	30.8	29.3	26.7	21.0	15.6
		MCWB	9.2	9.7	11.7	14.1	16.5	18.4	19.6	20.2	19.1	17.2	13.5	10.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.9	15.8	19.0	20.6	21.8	24.4	25.1	24.2	23.3	22.3	18.7	15.3
		MCDB	16.2	20.7	26.5	27.8	30.1	31.4	31.3	30.0	28.5	28.0	24.7	20.0
	2%	WB	11.4	13.0	16.2	18.1	19.7	22.0	23.3	23.1	22.0	20.7	17.1	13.5
		MCDB	14.6	17.3	21.8	26.9	28.0	27.9	28.5	28.4	27.0	25.6	21.9	17.0
	5%	WB	10.5	11.7	13.9	16.2	18.5	20.5	22.2	22.4	21.2	19.7	15.9	12.4
		MCDB	13.5	15.6	20.2	24.4	27.2	27.0	27.3	27.4	25.8	24.5	20.0	15.5
	10%	WB	9.8	10.6	12.5	14.9	17.4	19.5	21.4	21.9	20.6	18.9	15.0	11.4
		MCDB	12.5	14.2	18.1	23.0	25.5	26.4	26.4	26.6	25.1	23.7	18.5	14.3
Mean Daily Temperature Range	5% DB	MDBR	7.2	7.6	8.5	10.0	11.1	11.1	10.3	10.3	10.0	9.4	8.7	7.6
		MCDBR	10.0	11.2	12.1	13.5	13.6	12.5	11.2	11.3	11.7	11.8	11.3	10.0
		MCWBR	5.5	6.1	6.4	6.4	6.0	5.6	5.0	4.7	4.9	5.6	6.2	5.4
	5% WB	MCDBR	8.0	9.4	10.9	12.3	11.7	11.0	10.3	10.5	10.0	9.8	9.0	7.7
		MCWBR	5.3	6.0	6.9	6.7	6.1	5.5	4.7	4.5	4.5	5.4	5.6	4.9
Clear-Sky Solar Irradiance	taub	0.346	0.373	0.412	0.451	0.450	0.386	0.395	0.401	0.391	0.408	0.365	0.349	
	taud	2.325	2.231	2.104	1.999	2.032	2.255	2.240	2.224	2.255	2.190	2.300	2.330	
	Ebn at Noon	861	872	862	842	844	896	886	877	872	821	831	835	
	Edh at Noon	103	125	152	177	173	138	139	139	129	128	105	97	
All-Sky Solar Radiation	RadAvg	2.88	3.66	5.09	6.26	7.38	8.23	7.98	7.30	6.18	4.65	3.45	2.71	
	RadStd	0.20	0.34	0.33	0.37	0.33	0.14	0.09	0.12	0.16	0.20	0.19	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	+0.44	N/A	-1.73	N/A	N/A	N/A	N/A	-64	+158	+103

Nomenclature: See separate page